HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

SwasthTwin

A personalised Digital Twin that forecasts the risk of a hyperglycemia event in the next two hours by fusing historical EHR data with wearable and CGM streams

Team **Innovators**

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Research prototype · synthetic data only · not a medical device

The problem: high-glucose episodes are seen after they happen

Adults with or at risk of type 2 diabetes generate rich CGM and wearable data, but it is reviewed retrospectively

1

Data without foresight

CGM shows where glucose IS. Reports summarise where it WAS. Few tools estimate where it is heading for this specific person.

2

Context is split

The EHR (HbA1c, diagnosis, medicines) and the wearable stream (glucose, activity, sleep) sit in different systems and are rarely read together.

3

Clinician time is short

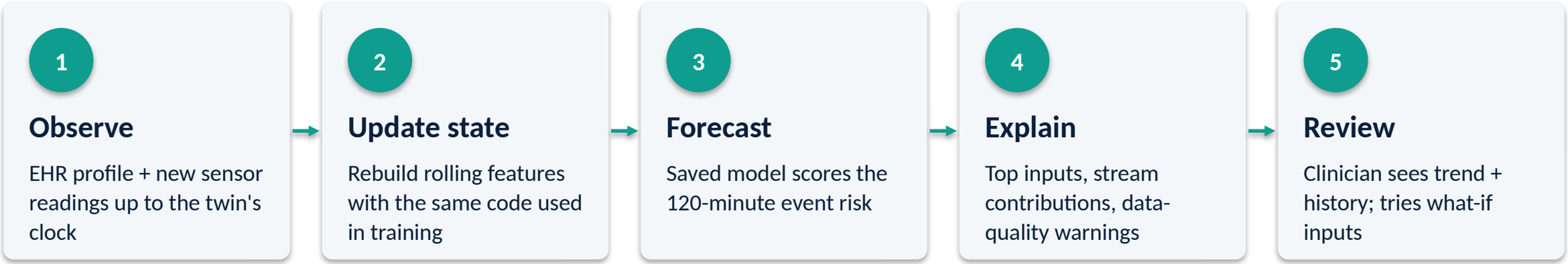
A care team cannot watch every stream. They need a short, explained signal: who may be heading high soon, and why the model thinks so.

Specific use case

Every 30 minutes, estimate P(any CGM reading ≥ 180 mg/dL in the next 120 minutes) for one patient, explain the estimate and flag data-quality problems — for clinician review, not for automatic action.

Why it is a Digital Twin, not just a prediction model

Each patient has a persistent virtual state that is updated as new observations arrive



What makes it a twin

- Patient-specific state object (EHR + observations + features + forecast history)
- Advances in time: deterministic replay of the sensor stream, or manual readings
- Every refresh is a real model call on updated inputs — no scripted percentages
- What-if mode edits a copy of the inputs and re-scores; the twin itself is unchanged

Honest scope

- Data-driven twin, not a mechanistic physiology model
- No live wearable or hospital integration — the feed is a replay
- What-if shows model sensitivity, not the body's response

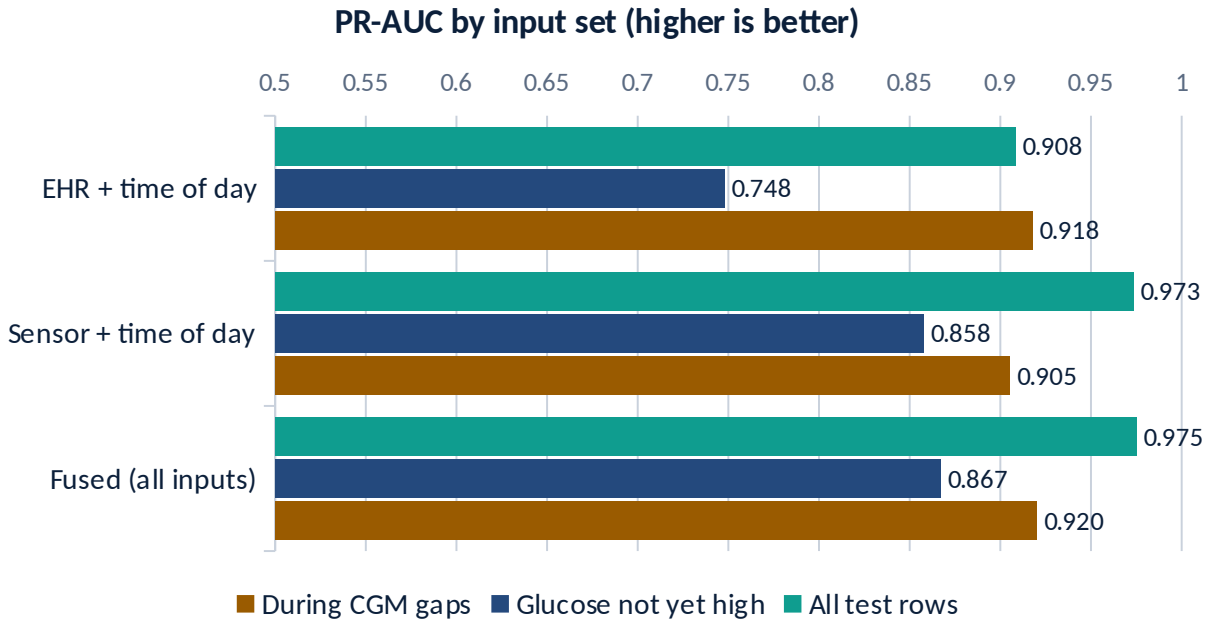
Two-stream fusion: sensors carry most signal, the EHR adds context

Same algorithm trained on different inputs, evaluated on held-out test patients

EHR · historical
Age, sex, BMI, diabetes status and duration, HbA1c, fasting glucose, eGFR, lipids, BP, medication class

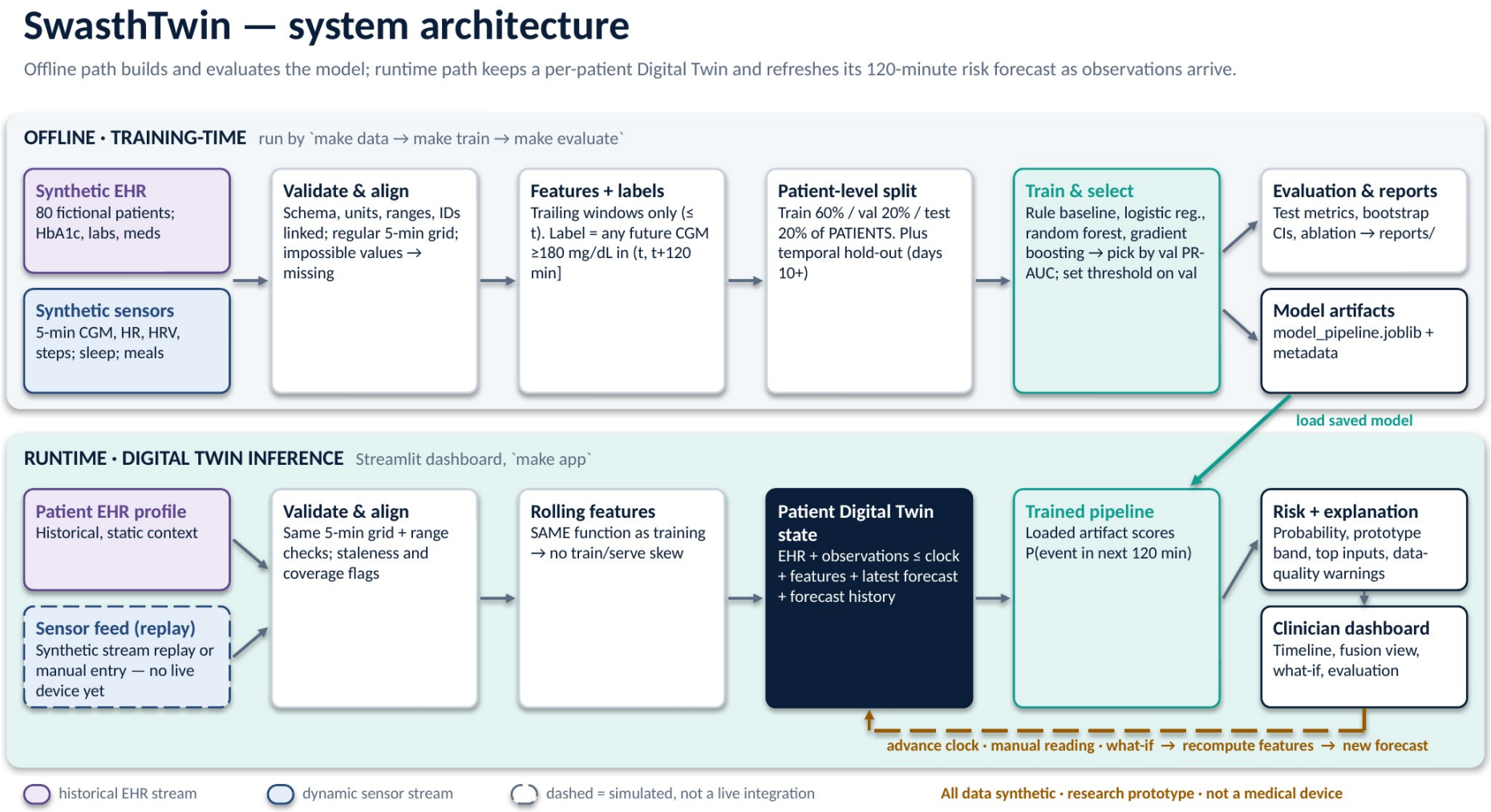
Sensors · dynamic
CGM level, trend, variability, coverage; heart rate, HRV vs own 24 h; steps; last night's sleep; optional meal log

Fusion features
Latest CGM minus the patient's EHR fasting glucose; HbA1c × logged carbs



- Adding the EHR to sensors: +0.002 PR-AUC overall, +0.015 during CGM gaps.
- Small gain; confidence intervals overlap (fused 0.939–0.988 vs sensor 0.938–0.986). We do not claim fusion is proven to help.

System architecture: offline training, runtime twin



Synthetic data, leakage-safe features, privacy by design

80 fictional patients × 14 days of 5-minute data, fully reproducible from one seed

Synthetic cohort

- EHR drives the simulator: higher HbA1c / BMI → larger meal spikes; meds damp them
- Meals, walks, stress, short sleep and dawn rise shape glucose; HR/HRV respond
- Realistic mess: CGM gaps, watch-off hours, noisy and incomplete meal logs
- Hand-picked coefficients — documented, not fitted to real data

Features (all from the past)

- CGM: level, 15/30-min trend, mean / SD / min / max, time in range, coverage
- Wearable: HR, HRV vs own 24 h, steps, active minutes; last night's sleep
- Meals count only from the moment they are logged
- Test: features identical whether or not future data exists

Privacy & safety

- No real person's data; IDs are SYN-0001…
- No names, phone numbers or government IDs anywhere
- Runs fully offline; no API keys or cloud services
- Prominent not-a-medical-device notice in every view

ML approach and leakage-resistant evaluation

Simple baselines first; choose the model on validation patients; touch test patients once

Rule baseline
No ML: alert if current or 60-min trend-projected CGM ≥ 180

Logistic regression
Linear baseline ML model

Random forest
Nonlinear ensemble

Gradient boosting (selected)
Histogram gradient-boosted trees

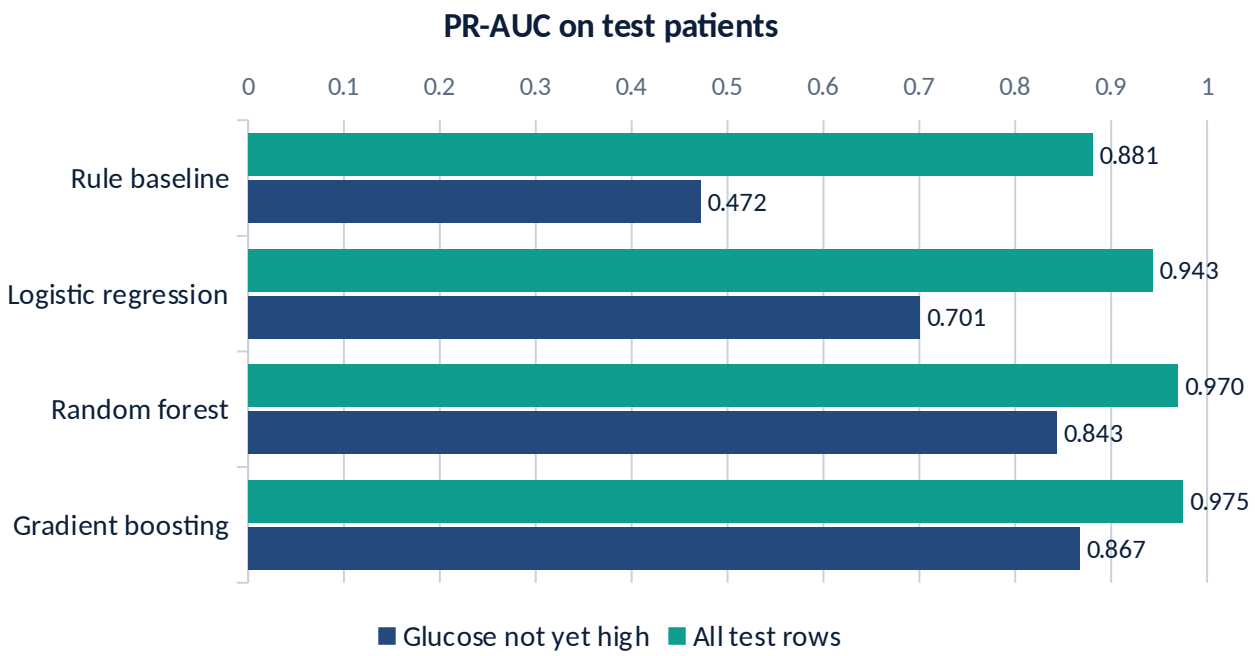
scikit-learn pipelines: median imputation + missing-value indicators, one-hot encoding, model. Saved with joblib with feature list, threshold and version.

Evaluation design

- **Split by patient:** 48 train / 16 validation / 16 test patients — nobody appears in two splits
- **Temporal check:** train on days 0–9 of training patients, test on days 10+ of test patients
- **Selection:** highest validation PR-AUC; alert threshold set on validation for ≥80% sensitivity
- **Uncertainty:** 95% CIs from a patient-level bootstrap (300 resamples)
- **Harder subset:** rows where glucose is not yet ≥ 180 — persistence alone cannot score well here
- **Also reported:** Brier score, calibration curve, confusion matrix, prevalence, permutation importance

Results on held-out synthetic patients

16 test patients · 10,275 forecasts · event prevalence 48% · all numbers from reports/metrics.json



0.975

Gradient boosting PR-AUC
95% CI 0.939–0.988

0.880

Rule baseline PR-AUC
95% CI 0.750–0.940

77%

sensitivity at threshold 0.86
(target was 80% on validation)

98%

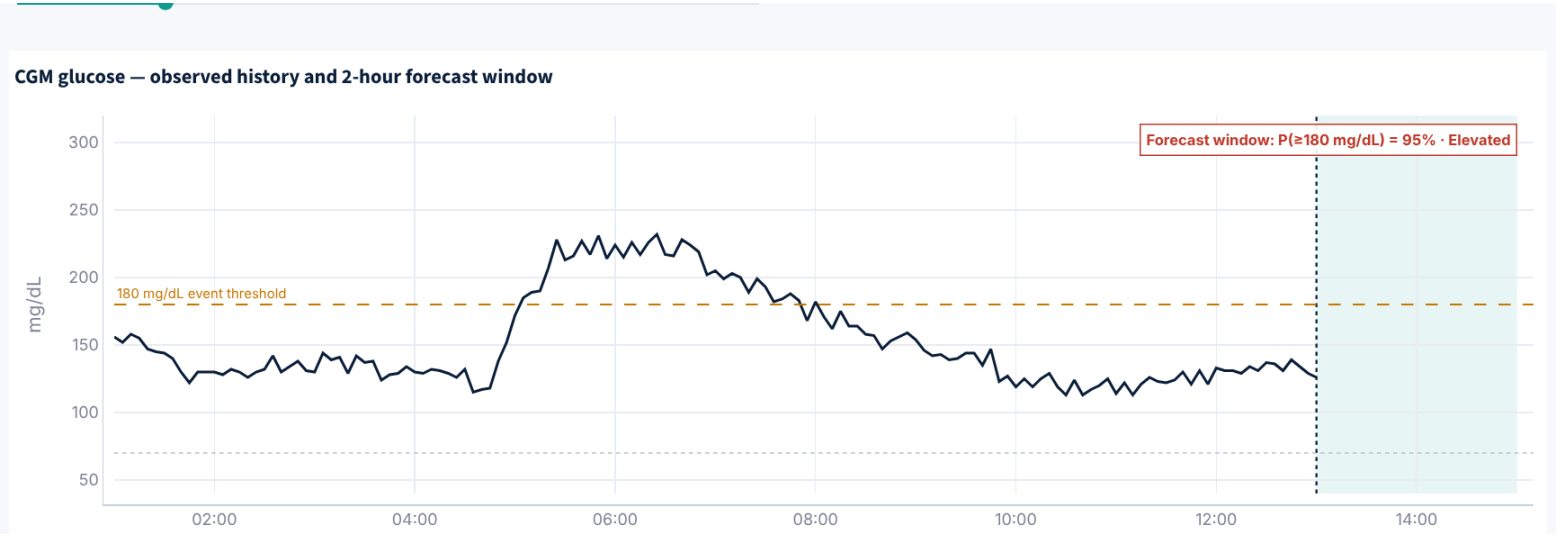
specificity · precision 98%

What this does and does not show

Where glucose is not yet high, PR-AUC 0.867 vs 0.472 for the rule. Temporal hold-out PR-AUC 0.976. Test sensitivity 77% fell short of the 80% validation target. Scores are high partly because the generator is learnable (e.g. regular meal times make clock time predictive). This is agreement with our simulator, NOT clinical evidence.

Live prototype: one synthetic patient, step by step

Held-out patient SYN-0002 · type2 diabetes · HbA1c 7.3% · values below produced by the saved model at build time



3 · What-if scenario (twin state is not changed)

Scenario

Add a brisk 30-minute walk (~480 steps per 5 min) to the last 30 minutes

BASELINE FORECAST	SCENARIO FORECAST	CHANGE
95%	62%	-33%
Elevated	Elevated	same model, edited inputs

Inputs that changed

feature	baseline	scenario
Heart rate, last 30 min (bpm)	85.3	110
Heart rate, last 120 min (bpm)	83	89.2
Steps, last 30 min	209	3,089
Steps, last 120 min	597	3,477
Active minutes, last 120 min	0	30

- Twin replayed 12 half-hour updates; its risk estimate ranged 2%–100% as the stream changed — every value a fresh model call.
- At Wed 13:00: latest CGM 126 mg/dL, P(event in 2 h) 95% (Elevated). The explanation shows time of day as a top driver — a known limitation of the synthetic routine.
- What-if (model sensitivity, not advice): +30-min walk → 62%; logging 80 g carbs now → 95%.

Potential value, limitations and the road to validation

Potential value (unproven)

- Earlier, explained flag for clinician review
- One place to read EHR + wearable context
- Data-quality warnings instead of silent guesses
- Reusable twin pattern for other forecasts

Key limitations

- Synthetic, hand-tuned simulator; results do not transfer
- 180 mg/dL / 2 h rule is a prototype target
- Clock time over-predictive due to regular synthetic meals
- Risk bands uncalibrated for clinical use
- No device or EHR integration; no hypoglycemia target

Roadmap

1. Public de-identified CGM data (licence checked) — retrain, re-test
2. Clinicians co-define target, threshold and alert burden
3. Retrospective study with ethics approval
4. Silent prospective pilot; calibration and safety review
5. Regulatory pathway before any clinical use

Ask: feedback from clinicians on the target definition and alert threshold, and access routes to consented real-world CGM + EHR data for validation.